Department of Mathematics
IIT Hyderabad
MA1130: Vector Calculus

Credits: 1

Fall 2026

Assignment 2

1. Evaluate $\displaystyle\int_C xy\,ds$, where C is the curve parametrized by $x = \cos t,\ y = \sin t,\ 0 \le t \le \frac{\pi}{2}$.

2. Evaluate $\displaystyle\int_C (2x + y)\,ds$, where C is the polygonal path from $(0, 0)$ to $(3, 0)$ and then to $(3, 2)$.

3. Evaluate $\displaystyle\int_C (x + y^2)\,ds$, where C is the path from $(2, 0)$ counterclockwise along the circle $x^2 + y^2 = 4$ to $(-2, 0)$ and then back to $(2, 0)$ along the x-axis.

4. Use a line integral to find the lateral surface area of the part of the cylinder $x^2 + y^2 = 4$ below the plane $x + 2y + z = 6$ and above the xy-plane.

5. Evaluate the line integral $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y) = (x^2 - y)\mathbf{i} + (x - y^2)\mathbf{j},$$

and C is the curve parametrized by

$$x = \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi.$$

6. Evaluate the line integral $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y) = xy^2\mathbf{i} + xy^3\mathbf{j},$$

and C is the polygonal path from $(0, 0)$ to $(1, 0)$ to $(0, 1)$ and then back to $(0, 0)$.

7. Evaluate the line integral $\displaystyle\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$$\mathbf{F}(x, y) = (x^2 + y^2)\mathbf{i},$$

and C is the curve parametrized by

$$x = 2 + \cos t, \quad y = \sin t, \quad 0 \le t \le 2\pi.$$

8. Evaluate $\displaystyle\oint_C (x^2 + y^2)\,dx + 2xy\,dy$, where C is parametrized by $x = \cos t,\ y = \sin t,\ 0 \le t \le 2\pi$.

9. Determine whether there exists a potential function $F(x, y)$ for the vector field $\mathbf{f}(x, y) = y\mathbf{i} - x\mathbf{j}$. If so, find one.

10. Determine whether there exists a potential function $F(x, y)$ for the vector field $\mathbf{f}(x, y) = xy^2\mathbf{i} + x^3\mathbf{j}$. If so, find one.

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11. Let $f(x,y)$ and $g(x,y)$ be continuously differentiable real-valued functions in a region R. Show that

$$\oint_C f\nabla g \cdot d\mathbf{r} = -\oint_C g\nabla f \cdot d\mathbf{r}$$

for any closed curve C in R.

12. Use Green's theorem to evaluate

$$\oint_C (x^2 - y^2)\,dx + 2xy\,dy,$$

where C is the positively oriented boundary of

$$R = \{(x,y) : 0 \le x \le 1,\ 2x^2 \le y \le 2x\}.$$

13. Use Green's theorem to evaluate

$$\oint_C 2y\,dx - 3x\,dy,$$

where C is the positively oriented circle $x^2 + y^2 = 1$.

14. Use Green's theorem to evaluate

$$\oint_C (e^{x^2} + y^2)\,dx + (e^{y^2} + x^2)\,dy,$$

where C is the positively oriented boundary of the triangle with vertices $(0,0)$, $(4,0)$ and $(0,4)$.

15. Determine whether there exists a potential function $F(x,y)$ for the vector field

$$\mathbf{f}(x,y) = (x^3\cos(xy) + 2x\sin(xy))\mathbf{i} + x^2 y\cos(xy)\mathbf{j}.$$

If so, find one.

16. Evaluate

$$\oint_C e^x \sin y\,dx + (y^3 + e^x\cos y)\,dy,$$

where C is the boundary of the rectangle with vertices $(1,-1)$, $(1,1)$, $(-1,1)$ and $(-1,-1)$, traversed counterclockwise.

17. For a region R bounded by a simple closed curve C, show that the area A of R is

$$A = -\oint_C y\,dx = \oint_C x\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx),$$

where C is traversed so that R is always on the left.

18. Use the Divergence Theorem to evaluate the surface integral

$$\iint_\Sigma \mathbf{f}\cdot d\boldsymbol{\sigma},$$

where $\mathbf{f}(x,y,z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and Σ is the boundary of the solid cube

$$S = \{(x,y,z) : 0 \le x,y,z \le 1\}.$$

19. Use the Divergence Theorem to evaluate the surface integral

$$\iint_\Sigma \mathbf{f}\cdot d\boldsymbol{\sigma},$$

where $\mathbf{f}(x,y,z) = x^3\mathbf{i} + y^3\mathbf{j} + z^3\mathbf{k}$ and Σ is the sphere

$$x^2 + y^2 + z^2 = 1.$$

20. Evaluate $\displaystyle\int_C f(x,y,z)\,ds$, where $f(x,y,z)=z$ and C is parametrized by

$$x=\cos t,\quad y=\sin t,\quad z=t,\quad 0\le t\le 2\pi.$$

21. Evaluate $\displaystyle\int_C f(x,y,z)\,ds$, where

$$f(x,y,z)=\frac{x}{y}+y+2yz,$$

and C is parametrized by

$$x=t^2,\quad y=t,\quad z=1,\quad 1\le t\le 2.$$

22. Evaluate the line integral $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F}(x,y,z)=(y-2z)\mathbf{i}+xy\mathbf{j}+(2xz+y)\mathbf{k},$$

and C is parametrized by

$$x=t,\quad y=2t,\quad z=t^2-1,\quad 0\le t\le 1.$$

23. Evaluate the line integral $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F}(x,y,z)=yz\mathbf{i}+xz\mathbf{j}+xy\mathbf{k},$$

and C is the polygonal path from $(0,0,0)$ to $(1,0,0)$ and then to $(1,2,0)$.

24. Evaluate the line integral $\displaystyle\int_C \mathbf{F}\cdot d\mathbf{r}$, where

$$\mathbf{F}(x,y,z)=xy\mathbf{i}+(z-x)\mathbf{j}+2yz\mathbf{k},$$

and C is the polygonal path from $(0,0,0)$ to $(1,0,0)$ to $(1,2,0)$ and then to $(1,2,-2)$.

25. Verify Stokes' theorem for the vector field

$$\mathbf{F}(x,y,z)=2y\mathbf{i}-x\mathbf{j}+z\mathbf{k},$$

where Σ is the upper hemisphere

$$x^2+y^2+z^2=1,\quad z\ge 0.$$

26. Verify Stokes' theorem for the vector field

$$\mathbf{F}(x,y,z)=xy\mathbf{i}+xz\mathbf{j}+yz\mathbf{k},$$

where Σ is the portion of the paraboloid

$$z=x^2+y^2,\quad z\le 1.$$

27. Let $\mathbf{F}=x^2\mathbf{i}-2xy\mathbf{j}+3xz\mathbf{k}$, and let D be the region in the first octant bounded by the sphere $x^2+y^2+z^2=4$. Use the Divergence Theorem to calculate the outward flux of $\mathbf{F}$ across the boundary of D.

28. Use Stokes' theorem to calculate the flux of the curl of the vector field $\mathbf{F}=2z\mathbf{i}+3x\mathbf{j}+5y\mathbf{k}$ across the surface S in the direction of the outward unit normal $\mathbf{n}$, where

$$S:\ \mathbf{r}(r,\theta)=(r\cos\theta)\mathbf{i}+(r\sin\theta)\mathbf{j}+(4-r^2)\mathbf{k},\quad 0\le r\le 2,\quad 0\le\theta\le 2\pi.$$

29. State and prove Green's theorem for a region with two holes.

30. Show that Green's Theorem is a special case of Stokes' Theorem.